Christopher Capobianco, MSc

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Skills

Programming Languages: Golang, C++, C, FORTRAN, R, Matlab
Scripting Languages: Python, Node, Bash, VBScript
Machine Learning: Keras, PyTorch, TensorFlow, Scikit-learn, NLP, Pandas, Matplotlib
Databases: PostgreSQL, MySQL, MariaDB, SQLite, Redis, DynamoDB
Technologies: AWS, GCP, JSON, XML, Regular Expressions, OpenMP, MPI, OpenAPI, OCR, Streamlit, Bitcoin
Tools: Docker, Kubernetes, RabbitMQ, Kafka, Git, Grafana, Prometheus, Metabase, Flask, FastAPI, Django, Express
Concepts: Server-Client, Micro-services, Distributed Systems, Object-Oriented Design, Parallelism
Mathematical & Numerical Techniques: Differential Equations, Fourier Analysis, Root Finding, Minimization

Experience

Senior Software Engineer, Okta Software Canada, Inc. – Kitchener, ON — May 2025 – Present
- Added Active Directory functionality to Okta Privileged Access (OPA) Client
- Added support for Non-Human Identities and Workloads to OPA Platform

R & D Team Lead, Arctic Wolf Networks – Waterloo, ON — Jan 2022 – Aug 2024
- Managed a team of five software developers, led to completion on two major projects:

1. Modularized monolithic Agent client, making updates less arduous for customers
2. Extricated a tightly coupled library from the macOS Agent, unblocking company IPO

Member of Technical Staff, Arctic Wolf Networks – Waterloo, ON — May 2020 – Jan 2022
- Decreased vulnerability diagnosis by one week for 95% of customers, by collecting more data during scan
- Decreased Agent on-boarding time by 90%, by creating a Python script to handle bulk scan group management

Software Developer, Green Brick Labs – Kitchener, ON — May 2015 – Mar 2020
- Designed and built crypto-currency financial system in Node.js
- Integrated payment providers into a Golang micro-service web application

Manager, Bank of Nova Scotia – Toronto, ON — Oct 2014 – Apr 2015
- Consolidated execution of value at risk models in production
- Validated consolidated model agreed with existing risk models

Research Developer, Prylynx Corporation – Kitchener, ON — Apr 2012 – Oct 2014
- Created ML document classifier using SVM, achieving 85% accuracy with humans in the loop
- Created UI to organize classified documents, and permit updating incorrect classifications

Research Assistant, Queen's University – Kingston, ON — Sept 2006 – Aug 2011
- Published investigation modeling giant planet formation via planetesimal accretion in a gaseous disk
- Parallelized SWIFT numerical integrator, achieving linear speed up

Projects

ML Portfolio — huggingface.co/spaces/ccapo/portfolio
- Fine-tuned Meta's Llama 3.1 (8B) Instruct to respond as a Math Assistant
- Created a Random Forest model to predict a document's classification, using OCR text
- Created a three layer LSTM model to generate novel musical compositions

Education

St. Mary's University – MSc in Astronomy — 2006
University of Toronto – BSc in Astrophysics — 2002